---CHUNK---

SOURCE: MEPC.59(33).pdf
nor the foundation connections and stack connections .
1.5 This specification provides emission requirements in annex AI, and fire protection requirements in annex A2 . Provisions for incinerators integrated with heat recovery units and provisions for flue gas temperature are given in annex A3 and annex A4, respectively.
1.6 This standard may involve hazardous materials, operations, and equipment. This standard does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use, including possible port State limitations.
2 Definitions
2.1 Ship means a vessel of any type whatsoever operating in the marine environment and includes hydrofoil boats, air-cushioned vehicles, submersibles, floating craft and fixed or floating platforms.
2.2 Incinerator means shipboard facilities for incinerating solid wastes approximating in composition to household waste and liquid wastes arising from the operation of the ship, e.g., domestic waste, cargo-associated waste,

---CHUNK---

SOURCE: MEPC.59(33).pdf
2.2 Incinerator means shipboard facilities for incinerating solid wastes approximating in composition to household waste and liquid wastes arising from the operation of the ship, e.g., domestic waste, cargo-associated waste, maintenance waste, operational waste, cargo residues, and fishing gear, etc. These facilities may be designed to use or not to use the heat energy produced.
2.3 Garbage means all kinds of victual, domestic and operational waste excluding fresh fish and parts thereof, generated during normal operation of the ship as defined in Annex V to MAR POL 73/78.
2.4 Waste means useless, unneeded or superfluous matter which is to be discarded.
2.5 Food wastes are any spoiled or unspoiled victual substances, such as fruits, vegetables, dairy products, poultry, meat products, food scraps, food particles, and all other materials contaminated by such wastes, generated aboard ship, principally in the galley and dining areas.
W/9096D/ljt/EWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992
REVISED GUIDELINES FOR THE IMPLEMENTATION OF
ANNEX V OF MARPOL 73/78MEPC 33/20/Add.1
ANNEX 2
Page 5

---CHUNK---

SOURCE: MEPC.59(33).pdf
aboard ship, principally in the galley and dining areas.
W/9096D/ljt/EWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992

2.6 Plastic means a solid material which contains as an essential ingredient one or more synthetic organic high polymers and which is formed (shaped) during either manufacture of the polymer or the fabrication into a finished product by heat and/or pressure. Plastics have material properties ranging from hard and brittle to soft and elastic~ Plastics are used for a variety of marine purposes including, but not limited to, packaging (vapour-proof barriers, bottles, containers, liners), ship construction (fibreglass and laminated structures, siding, piping, insulation, flooring, carpets, fabrics, paints and finishes, adhesives, electrical and electronic components), disposable eating utensils and cups, bags, sheeting, floats, fishing nets, strapping bands, rope and line.
2.7 Domestic waste means all types of food wastes, sewage and wastes generated in the living spaces on board the ship.

---CHUNK---

SOURCE: MEPC.59(33).pdf
disposable eating utensils and cups, bags, sheeting, floats, fishing nets, strapping bands, rope and line.
2.7 Domestic waste means all types of food wastes, sewage and wastes generated in the living spaces on board the ship.
2.8 Cargo-associated waste means all materials which have become wastes as a result of use on board a ship for cargo stowage and handling. Cargo-associated waste includes but is not limited to dunnage, shoring pallets, lining and packing materials, plywood, paper, cardboard, wire, and steel strapping.
2.9 Maintenance waste means materials collected by the engine department and the deck department while maintaining and operating the vessel, such as soot, machinery deposits, scraped paint, deck sweeping, wiping wastes, oily rags, etc.
2.10 Operational wastes means all cargo-associated wastes and maintenance waste (including ash and clinkers), and cargo residues defined as garbage in 2.13.
2.11 Sludge oil means sludge from fuel and lubricating oil separators, waste lubricating oil from main and auxiliary machinery, waste oil from bilge water separators, drip trays, etc.

---CHUNK---

SOURCE: MEPC.59(33).pdf
2.13.
2.11 Sludge oil means sludge from fuel and lubricating oil separators, waste lubricating oil from main and auxiliary machinery, waste oil from bilge water separators, drip trays, etc.
2.12 Oily rags are rags which have been saturated with oil as controlled in Annex I to the Convention. Contaminated rags are rags which have been saturated with a substance defined as a harmful substance in the other Annexes to MARPOL 73/78.
2.13 Cargo residues for the purposes of this standard are defined as the remnants of any cargo material on board that cannot be placed in proper cargo

holds (loading excess and spillage) or which remains in cargo holds and elsewhere after unloading procedures are completed (unloading residual and spillage). However, cargo residues are expected to be in small quantities.
2.14 Fishing gear is defined as any physical device or part thereof or combination of items that may be placed on or in the water with the intended purpose of capturing, or controlling for subsequent capture, living marine or freshwater organisms.
W/9096D/ljt/EWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992
REVISED GUIDELINES FOR THE IMPLEMENTATION OF

---CHUNK---

SOURCE: MEPC.59(33).pdf
purpose of capturing, or controlling for subsequent capture, living marine or freshwater organisms.
W/9096D/ljt/EWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992
REVISED GUIDELINES FOR THE IMPLEMENTATION OF
ANNEX V OF MARPOL 73/78MEPC 33/20/Add.1
ANNEX 2
Page 6
3 Materials and manufacture:
3.1 The materials used in the individual parts of the incinerator are to be suitable for the intended application with respect to heat resistance, mechanical properties, oxidation, corrosion, etc., as in other auxiliary marine equipment.
3.2 Piping for fuel and sludge oil should be seamless steel of adequate strength and to the satisfaction of the Administration. Short lengths of steel, or annealed copper nickle, nickle copper, or copper pipe and tubing may be used at the burners. The use of nonmetallic materials for fuel lines is prohibited. Valves and fittings may be threaded in sizes up to and including 60 mm 0.0. (outside diameter), but threaded unions are not to be used on pressure lines in sizes 33 mm 0.0. (outside diameter) and over.
3.3 All rotating or moving mechanical and exposed electrical parts should be protected against accidental contact.

---CHUNK---

SOURCE: MEPC.59(33).pdf
pressure lines in sizes 33 mm 0.0. (outside diameter) and over.
3.3 All rotating or moving mechanical and exposed electrical parts should be protected against accidental contact.
3.4 Incinerator walls are to be protected with insulated fire bricks/refractory and a cooling system. Outside surface temperature of the incinerator casing being touched during normal operations should not exceed 20°C above ambient temperature.
3.5 Refractory should be resistant to thermal shocks and resistant to normal ship's vibration. The refractory design temperature should be equal to the combustion chamber design temperature plus 20~. (See 4.1)
3.6 Incinerating systems should be designed such that corrosion will be minimized on the inside of the systems.

3.7 In systems equipped for incinerating liquid wastes, safe ignition and maintenance of combustion must be ensured, e.g., by a supplementary burner.
3.8 The combustion chamber(s) should be designed for easy maintenance of all internal parts including the refractory and insulation.
3.9 The combustion process should take place under negative pressure which

---CHUNK---

SOURCE: MEPC.59(33).pdf
3.8 The combustion chamber(s) should be designed for easy maintenance of all internal parts including the refractory and insulation.
3.9 The combustion process should take place under negative pressure which means that the pressure in the furnace under all circumstances should be lower than the ambient pressure in the room where the incinerator is installed. A flue gas fan may be fitted to secure negative pressure.
3.10 The incinerating furnace may be charged with solid waste either by hand or automatically. In every case, fire dangers should be avoided and charging should be possible without danger to the operating personnel.
For instance, where charging is carried out by hand, a charging lock may be provided which ensures that the charging space is isolated from the fire box as long as the filling hatch is open.
Where charging is not effected through a charging lock, an interlock should be installed to prevent the charging door from opening while the